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International Mathematics Assessments for Schools (2022-2023)

Upper Primary Division Round 2

Time: 120 minutes

Printed Name _____ *Code* _____ *Score* _____

Instructions:

- Do not open the contest booklet until you are told to do so.
- Be sure that your name and code are written on the space provided above.
- Round 2 of IMAS is composed of three parts; the total score is 100 marks.
- Questions 1 to 5 are given as a multiple-choice test. Each question has five possible options marked as A, B, C, D and E. Only one of these options is correct. After making your choice, fill in the appropriate letter in the space provided. Each correct answer is worth 4 marks. There is no penalty for an incorrect answer.
- Questions 6 to 13 are a short answer test. Only Arabic numerals are accepted; using other written text will not be honored or credited. Some questions have more than one answer, as such all answers are required to be written down in the space provided to obtain full marks. Each correct answer is worth 5 marks. There is no penalty for incorrect answers.
- Questions 14 and 15 require a detailed solution or process in which 20 marks are to be awarded to a completely written solution. Partial marks may be given to an incomplete presentation. There is no penalty for an incorrect answer.
- Use of electronic computing devices is not allowed.
- Only pencil, blue or black ball-pens may be used to write your solution or answer.
- Diagrams are not drawn to scale. They are intended as aids only.
- After the contest the invigilator will collect the contest paper.

The following area is to be filled in by the judges;
the contestants are not supposed to mark anything here.

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Total Score	Signature
Score																	
Score																	

Upper Primary Division Round 2

Questions 1 to 5, 4 marks each

1. Each digit of a 2023-digit number is 2. When this number is divided by 9, what is the remainder?

（A）3　　（B）4　　（C）5　　（D）6　　（E）7

Answer：_____

2. Alice and Bob were each paying $13000 a month for their respective one-bedroom apartments until the end of 2022. Starting in 2023, Alice's will go up by $100 every month, while Bob's rent will go up $500 every January and July. Starting from January 2023, when will be the month where the total accumulated rent paid by Alice will exceed the total accumulated rent paid by Bob?

（A）November 2024　　（B）December 2024　　（C）January 2025
（D）February 2025　　（E）March 2025

Answer：_____

3. A conveyor belt connects two platforms of a subway. If we are just standing still on the belt, the journey from one end of the belt to the other end lasts 120 seconds. Walking normally at a constant speed, outside the conveyor belt, the journey from one end of the belt to the other end lasts 180 seconds. How many seconds would it take to travel from one end to the other end of the conveyor belt in the same direction if you walk normally on the conveyor belt, with the same speed at which we walked outside to the conveyor belt?

（A）54　　（B）63　　（C）72　　（D）90　　（E）150

Answer：_____

4. Consider all four-digit numbers formed from one each of 0, 1, 2 and 3. How many numbers of those not beginning in 0 or not ending in 1 are there?

（A）12　　（B）13　　（C）14　　（D）15　　（E）16

Answer：_____

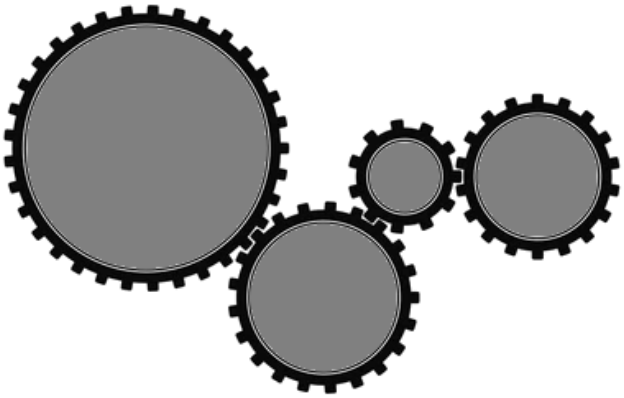
5. The numbers 90, 92, 94, 96 and 98 are written on five cards. A student draws one of them at random, and then she writes down three numbers. The first number written can be chosen arbitrarily by the student except that it must be larger than the number on her card. The second number is the difference between the first number and the number on her card. The third number is the sum of the first number and the number on her card. For which number on her card is it impossible for three prime numbers to be written down?

（A）90　　（B）92　　（C）94　　（D）96　　（E）98

Answer：_____

Questions 6 to 13, 5 marks each

6. Four cog wheels are in constant mesh. The largest cog has 650 teeth, the next 425 teeth, the next 300 teeth and the smallest has 150 teeth. If they start revolving now, how many revolutions will the largest cog have made before they are all turn back to their original positions at the same time for the first time? (The diagram below is an example of a cog set)

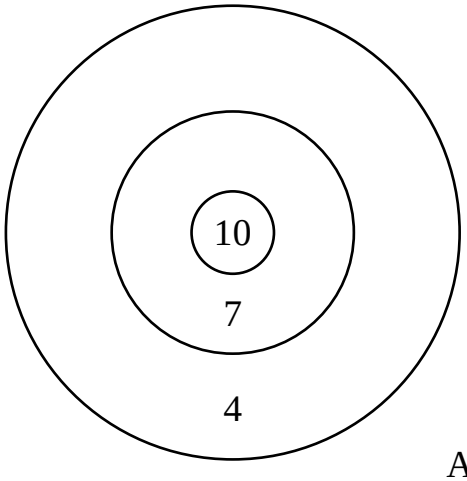


Answer：_____ revolutions

7. The finalists of a singing contest are Alice, Bob and Chris. Each judge voted for exactly one of the three participants. According to Dennis, there were 59 judges, with 30 voting for Alice or Bob, 13 voting for Bob or Chris, and 10 voting for Chris or Alice. However, Dennis has poor eyesight and each of the four counts which he provided could be off by up to 13. Exactly how many judges voted for Alice?

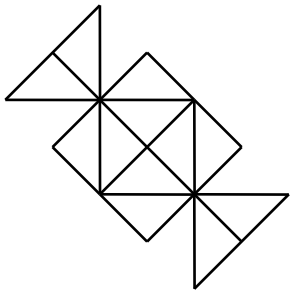
Answer：_____ judges

8. In the target board shown below, the centre is worth 10 points, the inner rim 7 points and the outer rim 4 points. A trainee hits the centre and the inner rim the same number of times, and misses the target once for every three shots. If the number of shots is a multiple of 3 and his total score is 99 points, what is the total number of shots he has thrown?

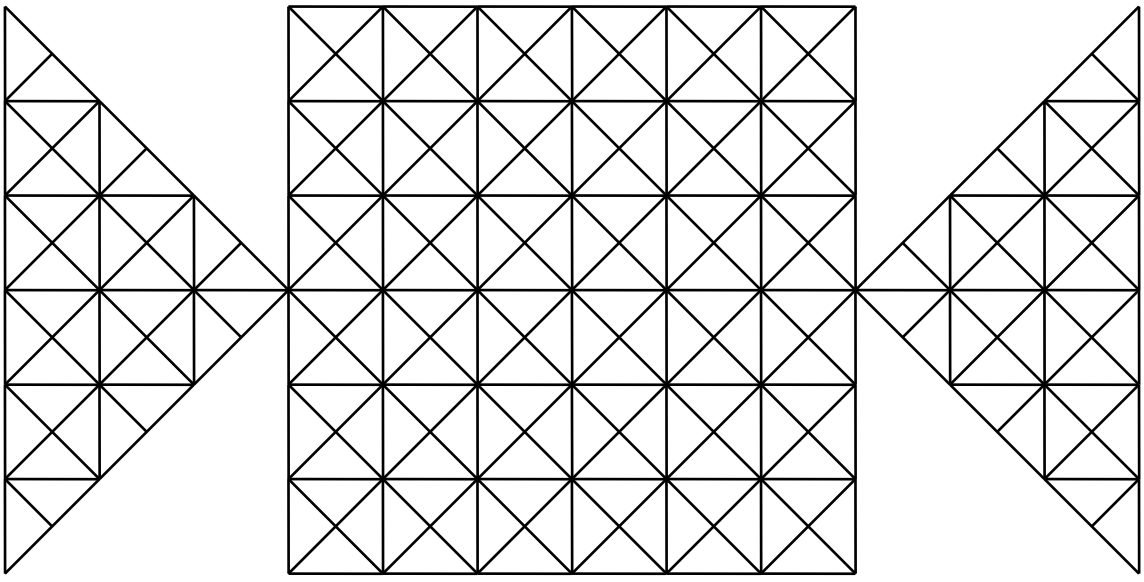


Answer : _____ shots

9. The diagram below shows a candy shaped figure:

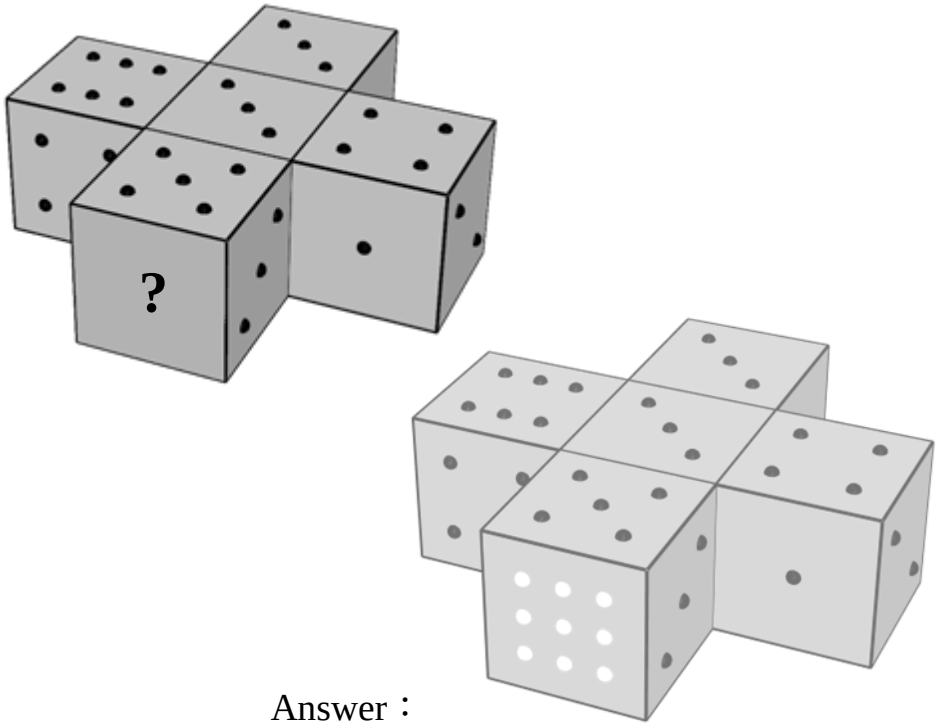


How many candy shapes with the same size as the one above can you find within the large grid below?



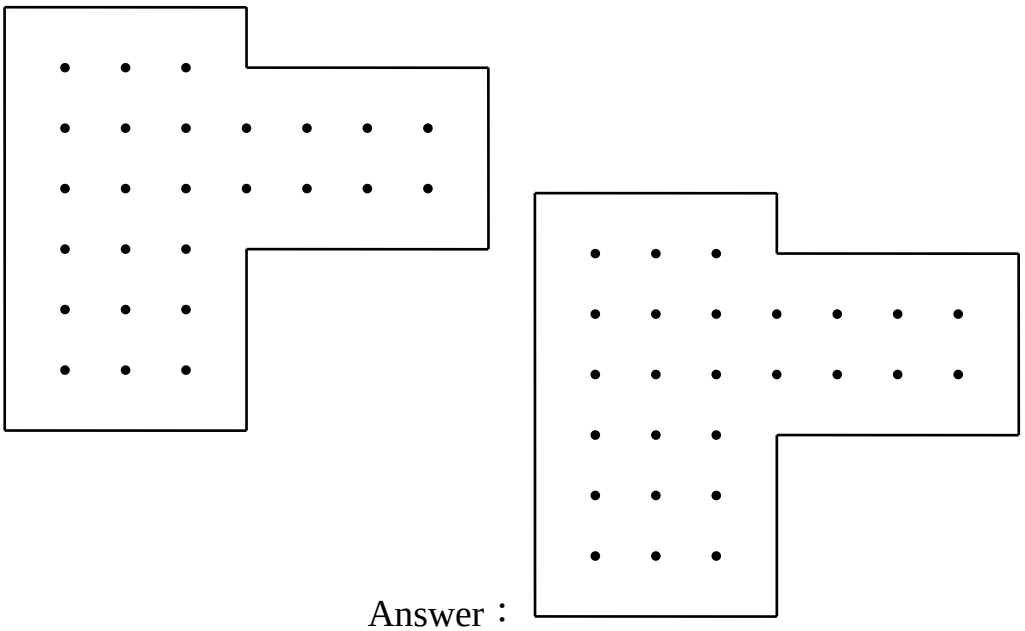
Answer : _____ shapes

10. Five standard dice in the cross below are exactly identical, including the orientation of dots on their faces. The numbers on opposite faces always add up to 7. Dice touch each other with the same numbers so that all dots on the touching faces exactly match. In other words, each of these dots touches exactly to another dot. Draw the position of these dots with exact orientation on the face with the question mark.



Answer：_____

11. Dissect the following diagram into two figures of the same size and the same shape after reflection and rotation.



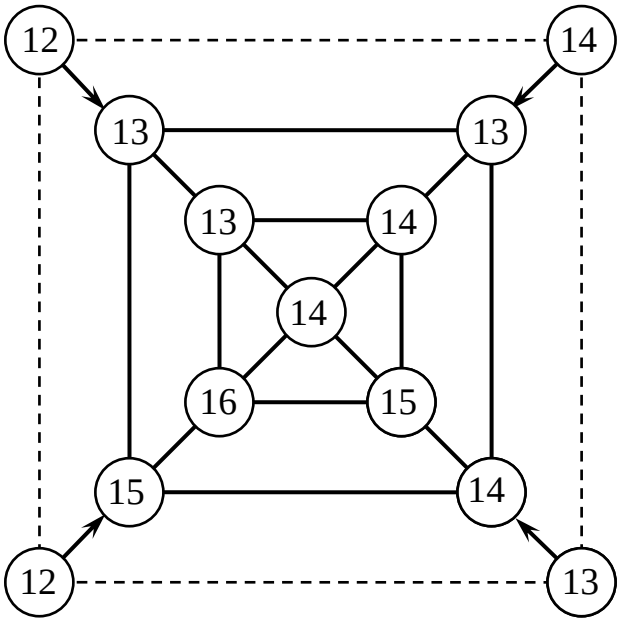
Answer：_____

12. Anna chooses some number of digits. Bob then chooses a positive integer and multiplies it by 3. Anna wins if at least one of her digits appears in either of Bob's two numbers. What is the minimum number of digits Anna must choose to guarantee that she wins no matter what number Bob chooses?

Answer： _____ digits

13. In the following map, Andy has to start at any one of the four corners and follow the other segments to the other four circles by following these instructions:
    - He passes through each segment at most once.
    - Each circle contains a pebble. Andy has to pick up the pebble when he enters a circle. The number in each circle represents the weight of the pebble in it, in g.
    - Andy cannot pass through more than one corner.

    How many different routes can Andy have to collect exactly 67 g of pebbles?



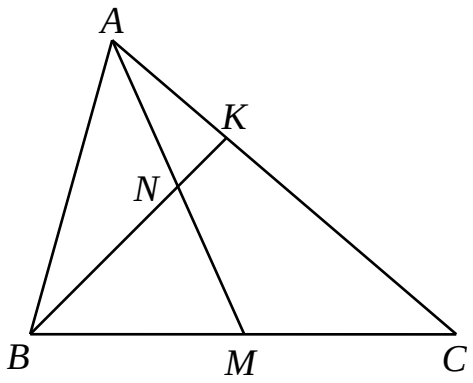
Answer： _____ routes

Questions 14 to 15, 20 marks each
(Detailed solutions are needed for these two problems)

14. A Mathematics test consists of 3 problems, where each problem is being graded independently with integer points between 0 and 10. How many ways can the total points for this test be exactly 21?

Answer : _____ ways

15. In triangle ABC, M is the midpoint of BC and N is the midpoint of AM. The extension of BN intersects AC at K, as shown in the figure below.
If $KA = 24$ cm, then what is the length, in cm, of AC?



Answer : _____ cm